

```
import pandas as pd

df = pd.read_csv('/content/credit_card_fraud_2025 (1).csv')
print(df.head())
```

	Transaction_ID	Customer_ID	Transaction_Date	Amount	Merchant_Category	
0	1	25795	2025-05-28 11:54:36	81.53	Online Services	
1	2	10860	2024-09-11 20:26:12	52.19	Fuel	
2	3	86820	2024-11-02 12:39:23	27.70	Utilities	
3	4	64886	2024-10-08 21:58:08	9.80	Clothing	
4	5	16265	2024-05-25 20:01:21	178.06	Electronics	

	Merchant_ID	Card_Type	Transaction_Type	Country	Is_International	Is_Chip	
0	8459	Gold	POS	Germany	1	1	
1	8205	Debit	ATM	Germany	0	1	
2	8076	Gold	Online	USA	0	1	
3	3244	Gold	ATM	USA	0	1	
4	3675	Gold	ATM	Germany	0	1	

	Is_Pin_Used	Distance_From_Home	Hour_of_Day	Device_Type	Fraud_Flag	
0	0	1.61	11	Web	0	
1	0	15.77	20	Web	0	
2	0	9.19	12	Terminal	0	
3	0	9.42	21	Web	0	
4	1	1.32	20	Web	0	

```
#Filtering data
non_fraud = df[df['Fraud_Flag'] == 0]
fraud = df[df['Fraud_Flag'] == 1]
```

```
non_fraud.head()
```

	Transaction_ID	Customer_ID	Transaction_Date	Amount	Merchant_Category	Merchant_ID	Card_Type	Transaction_Type	Country	Is_International	Is_Chip	Is_Pin_Used	Distance_From_Home	Hour_of_Day
0	1	25795	2025-05-28 11:54:36	81.53	Online Services	8459	Gold	POS	Germany	1	1	0	1.61	11
1	2	10860	2024-09-11 20:26:12	52.19	Fuel	8205	Debit	ATM	Germany	0	1	0	15.77	20
2	3	86820	2024-11-02 12:39:23	27.70	Utilities	8076	Gold	Online	USA	0	1	0	9.19	12
3	4	64886	2024-10-08 21:58:08	9.80	Clothing	3244	Gold	ATM	USA	0	1	0	9.42	21
4	5	16265	2024-05-25 20:01:21	178.06	Electronics	3675	Gold	ATM	Germany	0	1	1	1.32	20

```
import numpy as np
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))

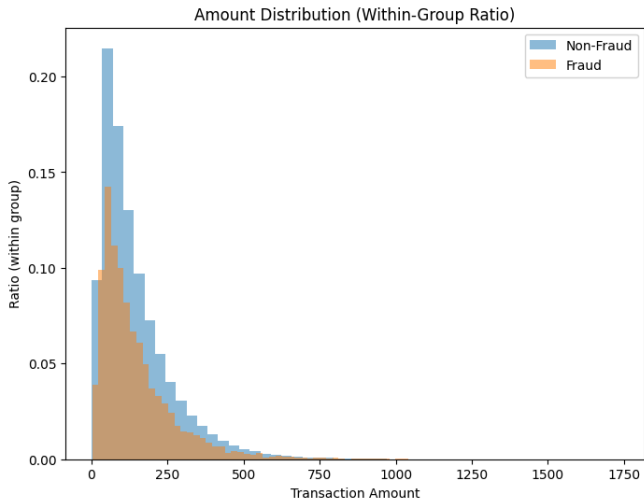
# normalize each subset
nf_weights = np.ones(len(non_fraud)) / len(non_fraud)
f_weights = np.ones(len(fraud)) / len(fraud)

plt.hist(non_fraud['Amount'], bins=50,
         weights=nf_weights, alpha=0.5, label='Non-Fraud')

plt.hist(fraud['Amount'], bins=50,
         weights=f_weights, alpha=0.5, label='Fraud')

plt.xlabel('Transaction Amount')
plt.ylabel('Ratio (within group)')
plt.title('Amount Distribution (Within-Group Ratio)')
plt.legend()

plt.show()
```



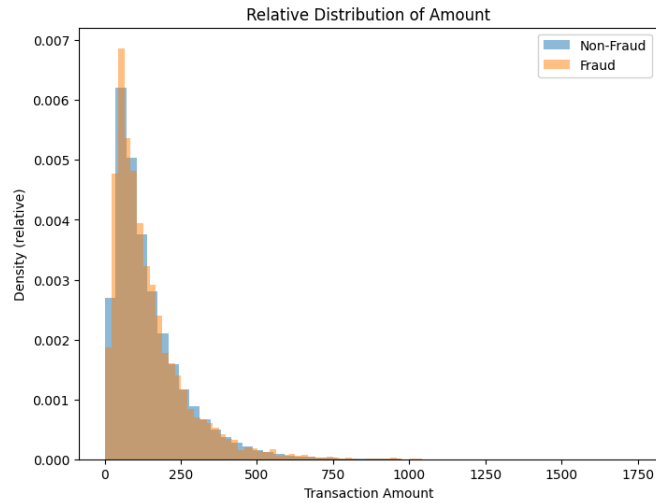
```
#Too much non-fraud data
plt.figure(figsize=(8,6))

plt.hist(non_fraud['Amount'], bins=50, alpha=0.5, density=True, label='Non-Fraud')
plt.hist(fraud['Amount'], bins=50, alpha=0.5, density=True, label='Fraud')

plt.xlabel('Transaction Amount')
```

```
plt.ylabel('Density (relative)')
plt.title('Relative Distribution of Amount')
plt.legend()

plt.show()
```



```
plt.figure(figsize=(8,6))

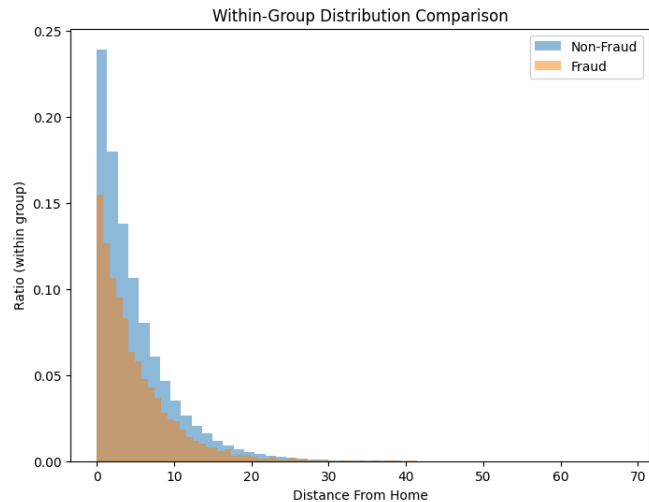
# normalize within each group
nf_weights = np.ones(len(non_fraud)) / len(non_fraud)
f_weights = np.ones(len(fraud)) / len(fraud)

plt.hist(non_fraud['Distance_From_Home'], bins=50,
         weights=nf_weights, alpha=0.5, label='Non-Fraud')

plt.hist(fraud['Distance_From_Home'], bins=50,
         weights=f_weights, alpha=0.5, label='Fraud')

plt.xlabel('Distance From Home')
plt.ylabel('Ratio (within group)')
plt.title('Within-Group Distribution Comparison')
plt.legend()

plt.show()
```



```
#Useless
import numpy as np
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))

nf_weights = np.ones(len(non_fraud)) / len(non_fraud)
f_weights = np.ones(len(fraud)) / len(fraud)

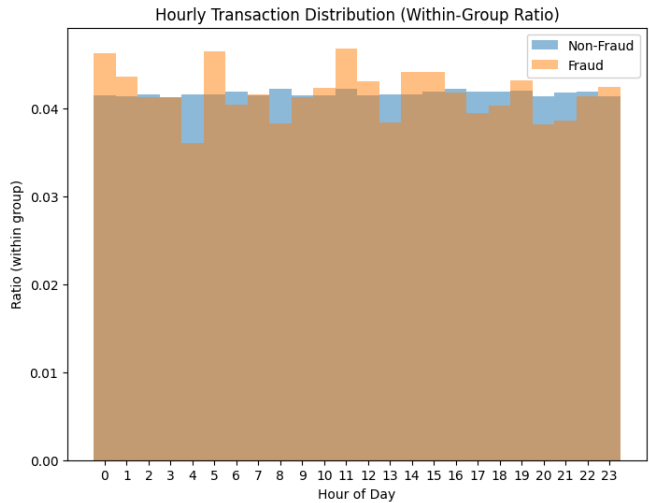
bins = np.arange(0, 25) - 0.5

plt.hist(non_fraud['Hour_of_Day'], bins=bins,
         weights=nf_weights, alpha=0.5, label='Non-Fraud')

plt.hist(fraud['Hour_of_Day'], bins=bins,
         weights=f_weights, alpha=0.5, label='Fraud')

plt.xlabel('Hour of Day')
plt.ylabel('Ratio (within group)')
plt.title('Hourly Transaction Distribution (Within-Group Ratio)')
plt.xticks(range(24))
```

```
plt.legend()
plt.show()
```



```
fraud.describe()
```

	Transaction_ID	Customer_ID	Amount	Merchant_ID	Is_International	Is_Chip	Is_Pin_Used	Distance_From_Home	Hour_of_Day	Fraud_Flag
count	7500.000000	7500.000000	7500.000000	7500.000000	7500.000000	7500.000000	7500.000000	7500.000000	7500.000000	7500.0
mean	251309.099067	54878.261467	145.095821	5489.690533	0.147600	0.300933	0.200933	4.914968	11.410133	1.0
std	145202.959473	25907.079117	123.940499	2619.700991	0.354727	0.458694	0.400725	4.816775	6.926975	0.0
min	13.000000	10013.000000	2.210000	1000.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.0
25%	124242.250000	32712.250000	59.965000	3215.750000	0.000000	0.000000	0.000000	1.420000	5.000000	1.0
50%	251159.000000	54837.000000	107.390000	5481.500000	0.000000	0.000000	0.000000	3.470000	11.000000	1.0
75%	376081.750000	77375.500000	188.237500	7752.250000	0.000000	1.000000	0.000000	6.900000	17.000000	1.0
max	499970.000000	99995.000000	1040.870000	9997.000000	1.000000	1.000000	1.000000	41.410000	23.000000	1.0

```
non_fraud.describe()
```

	Transaction_ID	Customer_ID	Amount	Merchant_ID	Is_International	Is_Chip	Is_Pin_Used	Distance_From_Home	Hour_of_Day	Fraud_Flag
count	492500.000000	492500.000000	492500.000000	492500.000000	492500.000000	492500.000000	492500.000000	492500.000000	492500.000000	492500.0
mean	249980.572095	55009.597232	145.270636	5501.401744	0.149155	0.300211	0.199783	5.003020	11.515269	0.0
std	144324.551738	25996.453983	120.835283	2597.146341	0.356242	0.458350	0.399837	5.001452	6.915091	0.0
min	1.000000	10000.000000	1.190000	1000.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.0
25%	125012.750000	32485.750000	60.810000	3255.000000	0.000000	0.000000	0.000000	1.430000	6.000000	0.0
50%	249978.500000	55060.000000	109.190000	5501.000000	0.000000	0.000000	0.000000	3.480000	12.000000	0.0
75%	374984.250000	77511.250000	192.290000	7751.000000	0.000000	1.000000	0.000000	6.920000	18.000000	0.0
max	500000.000000	99998.000000	1731.550000	9998.000000	1.000000	1.000000	1.000000	68.130000	23.000000	0.0

```
fraud_dist = fraud['Transaction_Type'].value_counts(normalize=True)
nonfraud_dist = non_fraud['Transaction_Type'].value_counts(normalize=True)

all_categories = sorted(df['Transaction_Type'].unique())
fraud_dist = fraud_dist.reindex(all_categories, fill_value=0)
nonfraud_dist = nonfraud_dist.reindex(all_categories, fill_value=0)

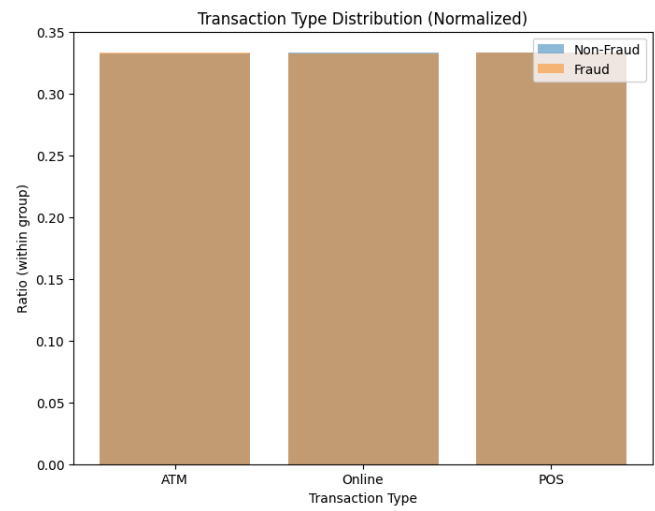
plt.figure(figsize=(8,6))

x = range(len(all_categories))

plt.bar(x, nonfraud_dist.values, alpha=0.5, label='Non-Fraud')
plt.bar(x, fraud_dist.values, alpha=0.5, label='Fraud')

plt.xticks(x, all_categories)
plt.xlabel('Transaction Type')
plt.ylabel('Ratio (within group)')
plt.title('Transaction Type Distribution (Normalized)')
plt.legend()

plt.show()
```



```
summary = df.groupby('Fraud_Flag')['Amount'].agg([
    'mean',
    'std',
    lambda x: x.quantile(0.25),
    lambda x: x.quantile(0.75)
])

summary.columns = ['Mean', 'Std', '25th Percentile', '75th Percentile']

print(summary)
```

Fraud_Flag	Mean	Std	25th Percentile	75th Percentile
0	145.270636	120.835283	60.810	192.2900
1	145.095821	123.940499	59.965	188.2375

```
from tabulate import tabulate

print(tabulate(summary, headers='keys', tablefmt='grid'))
```

Fraud_Flag	Mean	Std	25th Percentile	75th Percentile
0	145.271	120.835	60.81	192.29
1	145.096	123.94	59.965	188.237